Class 12

Chemistry

Unit : Solutions

Solutions

- Solutions are **homogeneous** mixtures of two or more than two components.
- The component that is present in the large quantity is known as **Solvent.**
- one or more components present in the solution other than the solvent are called **solutes.**

Types of solutions	state	solvent	Common examples
Gaseous solutions	Gas Liquid Solid	Gas Gas Gas	Mixture of oxygen and nitrogen gases. Chloroform mixed with nitrogen gas. Camphor in nitrogen gas.
Liquid solutions	Gas Liquid	Liquid Liquid	Oxygen dissolved in water. Ethanol dissolved in

	Solid	Liquid	water. Glucose dissolved in water.
Solid solutions	Gas Liquid Solid	Solid Solid solid	Solution of hydrogen in palladium. Amalgam of mercury with sodium. Copper dissolved in gold.

Terms used to define the concentration

1. Mass%(w/w)

Mass % of a component= $\frac{mass\ ooo\ the\ solute}{mass\ o00\ the\ solutsson} \times$

2. volume%(v/v)

volume % of a component= $\frac{volume\ ooo\ the\ component}{total\ volume\ o00\ solutsson}$

mass by volume%(w/v) is the mass of solute which is dissolved in 100ml of solution.

3. Parts per million(ppm)

Parts per million= $\frac{number\ o0\ 0\ parts\ o0\ 0\ the\ component}{total\ no\ o00\ parts\ o00\ all\ component\ o00\ the\ solutsson} \times 10^{6}$

It is the ratio of number of moles of one component to the total number of components present in the solution .

4. Mole fraction(x)

Mole fraction of the solute(x_{solute}) $\frac{n(solute)}{n(solute)+n(solvent)}$

5. Molarity(M)

It is defined as the number of moles of solute dissolved in one litre (or one cubic decimeter) of solution.

Molarity(M) $\frac{moles\ ooo\ solute}{volume\ o00\ solutsson\ ssn\ lstre}$

6. Molality(m)

It is defined as the number of moles of the solute per kg of the solvent.

Molality $\frac{moles\ ooo\ solute}{mass\ ooo\ solvent\ ssn\ kg}$

Solubility

Solubility is defined as the maximum amount of substance that can be dissolved in a specific amount of solvent at the particular temperature. It depends upon the nature of the solute and solvent at specified temperature and pressure.

Solubility of a solid in a liquid

Every solid does not dissolve in a given liquid.

Polar solute dissolve in polar solvent and non polar solute dissolve in non polar solvent, if the intermolecular interactions are same or we can say that **like dissolves like**.

Saturated solutions are those solutions in no more solute can be added dissolve at the same temperature and pressure for **unsaturated solutions** it is vice-versa.

- **There are two factors which affect the solubility of a solid in a liquid.**

  1. **Temperature**

     If dissolution of solid is endothermic ie,($\Delta H = -ve$).

     Then solubility decreases with increase in temperature.

     If it is exothermic ie, ($\Delta H = +ve$).

     Then solubility increases with decreases in temperature.

  2. **Pressure**

     It does not have any effect on solubility because solid and liquids are highly incompressible.

Solubility of a gas in a liquid

Many gases dissolve in a liquid.

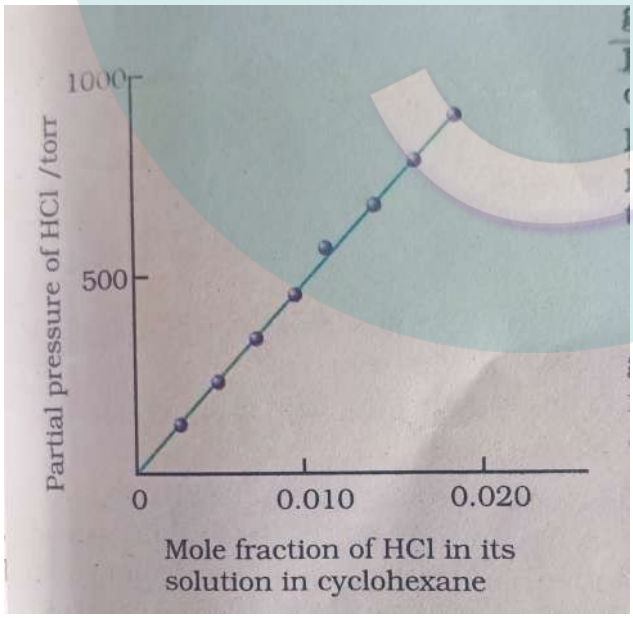
- **Factors affecting the solubility of a gas in a liquid**

1. **Temperature**

   Temperature increases the intermolecular attraction between the molecules becomes decreases hence solubility of gas in liquid decreases.

2. **Pressure**

   The solubility of gases increase with increase of pressure. **Henry** he was the first who gives the relationship between pressure and solubility of a gas in a solvent which is known as **Henry's Law.**



Henry's law states that at the constant temperature

"the partial pressure of the gas in a vapour phase(p) is proportional to the mole fraction of the gas(x) in the solution"

$$P=k_H X$$

K_H is the henry's constant.

different gases has different value of k_H

Higher the value of k_H at a given pressure, lower the solubility of gas in a liquid.

- **Applications of henry's law**

1. Soda cans or cold drink bottles are sealed at high pressure for increasing the solubility of co_2 gas in the solvent.
2. Scuba divers air tanks are diluted by addition of He to prevent formation of bends(11.7%He,56.2%N_2 and 32.1%o_2).

Aquatic animals are more comfortable in cold water rather then warm water because dissolving power of oxygen in water decreases with rising temperature and increases with lowering the temperature.

- **Limitations of henry's law**

1. For non ideal solutions this law is not applicable.

2. This law is only applicable for those gases which do not react with solvent.

Vapour pressure

Pressure exerted by vapour at the surface of the liquid at equllibrium.

Factors on which vapour pressure is depend.

1. **Temperature**

   Vapour pressure is directly proportional to the temperature.

   temptt v.p tt

2. **Intermolecular force of attraction**

   Vapour pressure $\propto \frac{1}{\text{Intermolecular force of attraction}}$

Roult's law

It states that for solution of volatile liquids, the partial vapour pressure of each component in the solution is directly proportional to its mole fraction.

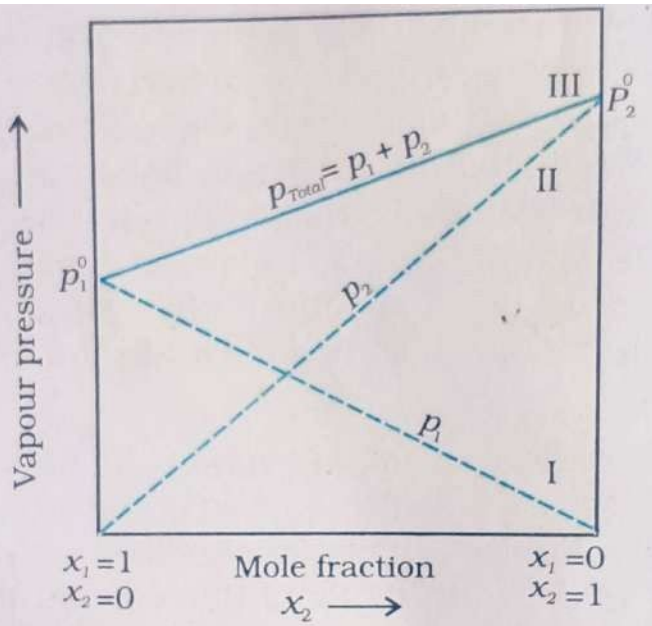


Fig. 2.3: *The plot of vapour pressure and mole fraction of an ideal solution at constant temperature. The dashed lines I and II represent the partial pressure of the components. (It can be seen from the plot that p_1 and p_2 are directly proportional to x_1 and x_2, respectively). The total vapour pressure is given by line marked III in the figure.*

P_1 a X_1	**P_1= partial pressure of 1**
$P_1 = P^\circ_1 X_1$	**P°_1= vapour pressure of pure 1**
P_2 a X_2	**P_2= partial pressure of 2**
$P_2 = P^\circ_2 X_2$	**P°_2= vapour pressure of pure 2**

According to **Dalton's law of partial pressure.** The total pressure of the solution phase in the container will be the sum of partial pressure of the the component of the solutions.

$$P_{total} = p_1 + p_2 \qquad (1)$$

Putting the value of p_1 & p_2 in (1)

$$P_{total} = x_1 P^{\circ}_1 + x_2 P^{\circ}_2$$

$$= (1-x_2) P^{\circ}_1 + x_2 P^{\circ}_2 \qquad (2)$$

$$= P^{\circ}_1 + (P^{\circ}_2 - P^{\circ}_1)x_2 \qquad (3)$$

If y_1 & y_2 are the mole fraction of component 1 & 2 in vapour phase then using Dalton's law of partial pressure.

$$P_1 = y_1 p_{total}$$

$$P_2 = y_2 p_{total}$$

$$P_1 = y_1 p_{total}$$

So mole fraction of component in vapour phase

$$Y_1 = \frac{p_1}{p_{total}}$$

Ideal solution

The solutions which obeys the Roult's law over the entire range of concentration.

In ideal solution

- The enthalpy of mixing of pure component to form the solution is zero and the volume of mixing is also zero.(means no heat is absorbed or evolved when two components are mixed and volume of solution would be equal to the sum of the volume of the components.)

Non ideal solution

The solution which do not obey the Roult's law over the entire range of concentration.

In non ideal solution

- The vapour pressure of such a solution is either higher or lower than that predicted by the Roult's law. If it is higher then the solution exhibits **positive deviation** if it is lower then the solutionexhibits **negative deviation**.

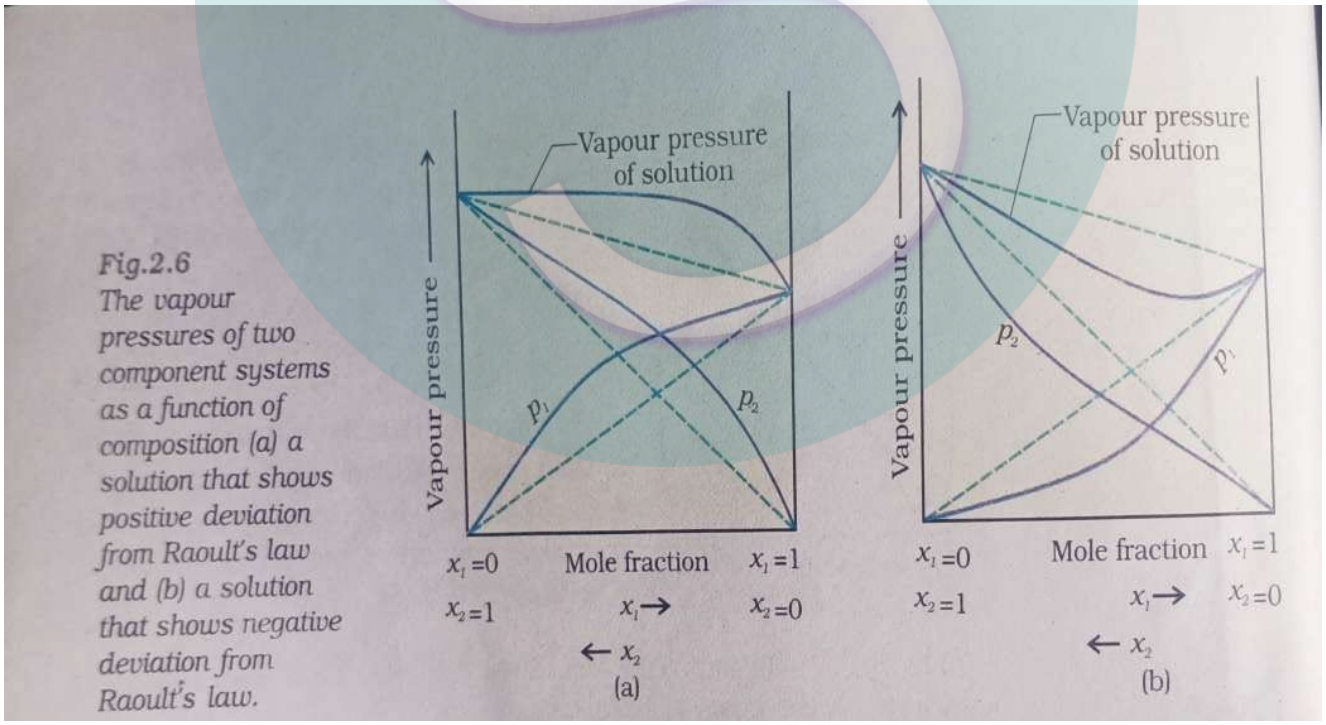


Fig.2.6 The vapour pressures of two component systems as a function of composition (a) a solution that shows positive deviation from Raoult's law and (b) a solution that shows negative deviation from Raoult's law.

Azeotropes

These are the mixture

There are two types of azeotropes

Minimum boiling azeotropes	**Maximum boiling azeotropes**
They are made from those liquid pairs which show positive deviation from ideal behavior. e.g., ethanol-water mixture.	They are made from those liquid pairs which show negative deviation from ideal behavior. e.g., nitric acid-water mixture.

Colligative properties

These are the property which depends only on the numbers of solute particles dissolve in a definite amount of the solvent not on the nature of the solute particles.

They are of four types

1. Relative lowering of vapour pressure

When a non volatile solute is dissolved in a solvent, vapour pressure of the solution is lower than that of the pure solvent which is known as lowering of vapour pressure.

Relative lowering of vapour pressure is equal to the mole fraction of the solute in the solution.

$$P_1 = x_1 p^\circ_1$$

The reduction in the vapour pressure pf solvent(Δp_1)is given as

$$\Delta p_1 = p^\circ_1 - p_1$$

$$= p^\circ_1 - x_1 p^\circ_1$$

$$= p^\circ_1 (1 - x_1)$$

$$\Delta p_1 = x_2 \, p^\circ_1$$

In a solution containing several non-volatile solutes, the lowering of the vapour pressure depends on the sum of the mole fraction of different solutes.

$$\frac{\Delta pppp1111}{pppp1111} = \frac{pppp^\circ 1111 - pppp^\circ 1111}{pppp^\circ 1111} = x_2$$

$$\frac{pppp^\circ 1111 - pppp1111}{pppp^\circ 1111} = \frac{nnnn \quad nn \quad n}{nnnn1111 + nnnnnnnn}$$

$$\frac{pppp^\circ 1111 - pppp1111}{pppp^\circ 1111}$$

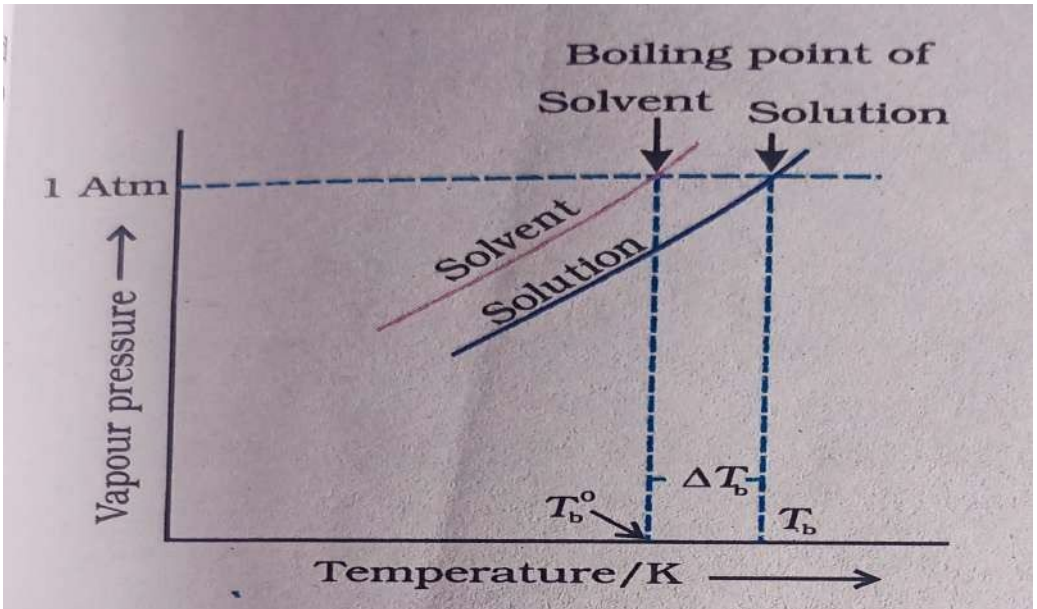
$$pppp^\circ 1111$$

$$\frac{pppp^\circ 1111 - pppp1111}{}$$

$$\frac{r_{2nnnn}}{r_{1111}} = \frac{W_{WWWnnnn} \times M_{1111}}{M_{nnnn} \times W_{WWWW1111}}$$

2. Elevation in boiling point

The boiling point of a solution containing a non volatile solute is always higher than the boiling point of the pure solvent. This increase in boiling point is termed as elevation in boiling point.



g. 2.7: The vapour pressure curve for solution lies below the curve for pure water. The diagram shows that ΔT_b denotes the elevation of boiling point of a solvent in solution.

$\Delta T_b = T_b - T^\circ_b$

ΔT_b a **m or** $\Delta T_b = K_b m$

$$= K_b\left(\frac{W_{WWW} \times 1000}{M \times W_{WWW}(ssn\ g)}\right)$$

$$\text{Or } M_2 = \frac{Kb \times W_{WWW} \times 1000}{\Delta T_b \times W}$$

K_b is called the boiling point elevation constant or molal elevation or (ebullioscopic constant) constant, having unit K kg mol^{-1}

3. Depression in freezing point

The freezing point of a solution containing a non volatile solute is always less than the freezing point is termed ass as depression in freezing point.

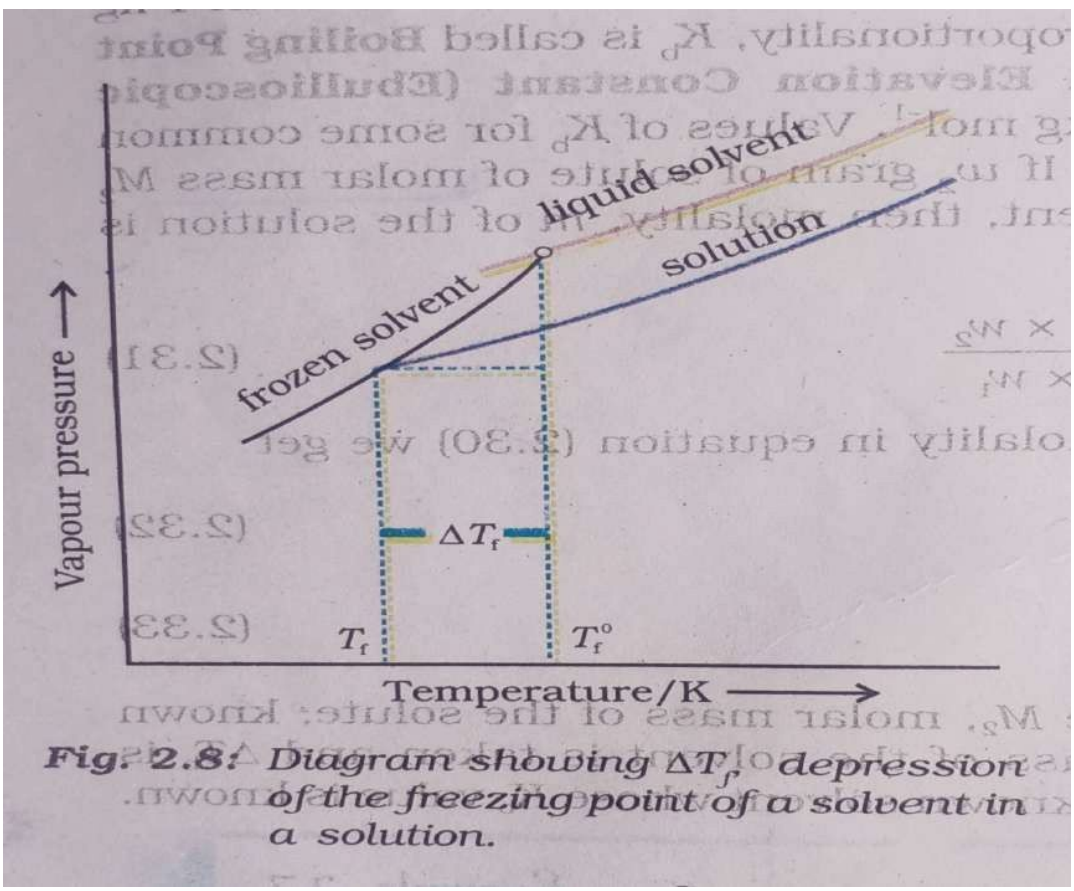


Fig. 2.8: Diagram showing ΔT_f, depression of the freezing point of a solvent in a solution.

$\Delta T_f = T^{\circ}_f - T_f$

ΔT_f a **m or** $\Delta T_f = k_f m$

$$= K_f\left(\frac{W_{WWW} \times 1000}{M \times W_{WWW}(ssn\ g)}\right)$$

$$\text{Or } M_2 = \frac{K_{00} \times W_{WWW} \times 1000}{\Delta T_f \times W_{1000}}$$

K_f is known as the freezing point depression constant or molal depression constant, having unit K $kgmol^{-1}$

The values of K_f & K_b which depends upon the nature of the solvent, can be ascertained from the following relations.

$$K_f = \frac{R \times M_{1000} \times T^2_{00}}{1000 \times \Delta_{vap}H} \qquad K_b = \frac{R \times M_{1000} \times T^2_{b}}{1000 \times \Delta_{vap}H}$$

Osmosis

When solvent molecules flows from lower concentration to higher concentration through semi permeable membrane.

Endoosmosis

When water enter in a cell through semi permeable membrane is called endoosmosis.

Exoosmosis

When water comes out from a cell through semi permeable membrane is called exoosmosis.

Osmotic pressure(π)

It is the pressure which stops the flow of solvent

π a **C** **(C=molarity)**

π = **CRT**

$\pi = \left(\frac{n}{v}\right)RT$

$\pi = \frac{WRT}{VM}$

Isotonic solution

Solution having same osmotic pressure at a given temperature is known as isotonic solution.

Solution with higher osmotic pressure is called **Hypertonic solution.**

Solution with lower osmotic pressure is called **Hpotonic solution.**

Reverse osmosis

When solvent moves from solution of higher concentration to solution of lower concentration through semi permeable membrane. this phenomenon is known ass reverse osmosis.

Abnormal molecular mass

When the molecular mass of the substance which is determined by any of the colligative properties comes out to be different from the expected value, then the substance is used to show abnormal molecular mass. This difference is shown when the solution is non ideal or the solute is undergoes association or dissociation.

Van't Hoff Factor

It is the ratio of the experimental value of the colligative property to the calculated value of the colligative property.

$$i = \frac{Normal\ molar\ mass}{Abnormal\ molar\ mass}$$

$$i = \frac{Observed\ collssgatssve\ property}{calculated\ collssgatssve\ property}$$

$$i = \frac{Number\ ooo\ moles\ ooo\ pertsscles\ aooter\ assocssatsson/dssssocssatsson}{Number\ ooo\ moles\ ooo\ partsscles\ beooore\ assocssatsson/dssssocssatsson}$$

van't hoff modifies the equation for the colligative properties

- **Relative lowering pressure** $\frac{p°1 - p1}{p} = i.\frac{n2}{n1}$
- **Elevation of boiling point** $\Delta T_b = i.K_b m$
- **Depression of freezing point** $\Delta T_f = i.\ K_f m$
- **Osmotic pressure of solution** $\pi = i.n_2RT/V$